Phanindra Dewan

Curriculum Vitae

Indian Institute of Science
Bengaluru, Karnataka, India - 560012
phanindrad@iisc.ac.in
phannyd.github.io

Education

2024– **Ph.D. in Physics**, *Indian Institute of Science*, Bengaluru
Advisor: Dr. Sumantra Sarkar

2022–2024 **MS in Physics**, *Indian Institute of Science*, Bengaluru, CGPA: 9.5/10
Project guide: Dr. Sumantra Sarkar

2019–2022 **B.Sc. (Hons.) in Physics**, *St. Xavier's College*, Kolkata, CGPA: 8.65/10

Research Interests

Fields Biological Physics, Soft Matter Physics, Statistical Physics, Active Matter

Skills Python, Mathematica, HPC/cluster computing, LaTeX

Publications

[1] Amrapali Datta, Phanindra Dewan, Aswin Anto Puthoor, Tanya Chhabra, Tanishq Tejaswi, Sindhu Muthukrishnan, Akshar Rao, Sumantra Sarkar, and Medhavi Vishwakarma. "Heterotypic interfacial tension between oncogenic and wild-type populations forms the mechanical basis of tissue-specific oncogenesis in epithelia". In: *eLife* 14 (Aug. 2026). ISSN: 2050-084X. DOI: `10.7554/elife.106893.4`. URL: `http://dx.doi.org/10.7554/eLife.106893.4`.

[2] Phanindra Dewan and Sumantra Sarkar. "Molecular communication in one-dimensional channels with active transport and crowding". In: *Phys. Rev. E* 113 (5 May 2026), p. 054105. DOI: `10.1103/mzhn-fc29`. URL: `https://link.aps.org/doi/10.1103/mzhn-fc29`.

[3] Soumyadeep Mondal, Phanindra Dewan, Lakshman Santhosh Kumar, and Sumantra Sarkar. "Symmetry-Protected Phases in a 1D Active Solid with Mechanochemical Feedback". In: *Phys. Rev. Lett.* 137 (7 Aug. 2026), p. 078401. DOI: `10.1103/p1bw-v87g`. URL: `https://link.aps.org/doi/10.1103/p1bw-v87g`.

[4] Sindhu Muthukrishnan, Phanindra Dewan, Tanishq Tejaswi, Michelle B. Sebastian, Tanya Chhabra, Soumyadeep Mondal, Soumitra Kolya, Sumantra Sarkar, and Medhavi Vishwakarma. "Glassy dynamics in active epithelia emerge from an interplay of mechanochemical feedback and crowding". In: *Nature Communications* 17.1 (2026). ISSN: 2041-1723. DOI: `10.1038/s41467-026-74163-0`. URL: `http://dx.doi.org/10.1038/s41467-026-74163-0`.

Collaborations

2024 – **Dr. Medhavi Vishwakarma's Epithelial Mechanobiology Lab**, *Department of Bioengineering, Indian Institute of Science*, Modelling tissue behaviour using vertex models with mechanochemical feedback

Talks

2026 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *3-minute short talk*, "Active, Adaptive, Autonomous Matter", WE-Hereaus Summer School, Institut d'Études Scientifiques de Cargèse, Cargèse, France.

2025 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", Discussion Meeting on Biological Physics and Mechanobiology, Department of Bioengineering, IISc, Bengaluru.

Posters Presented

2026 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *Complexity in Matter and Policy*, IISc, Bengaluru.

2026 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *"Active, Adaptive, Autonomous Matter"*, WE-Hereaus Summer School, Institut d'Études Scientifiques de Cargèse, Cargèse, France.

2026 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *International Soft Matter Conference (ISMC) 2026*, Goa, India.

2025 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *CompFlu 2025*, IISc, Bengaluru.

2025 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *Unifying Concepts in Glass Physics (UCGP) 2025*, RRI, Bengaluru.

2025 "Emergence of glassy dynamics in epithelia via interplay of mechanochemical feedback and crowding", *1st Indian Mechanobiology Meeting*, NCBS, Bengaluru.

2025 "Role of interfacial tension in the outcome of cell-competition during cancer initiation across tissues", *Frontiers in Non-Equilibrium Physics (FNEP) II*, IMSc, Chennai.

2024 "A potpourri of results on molecular communication with active transport", *In-House Symposium 2024*, Department of Physics, at IISc, Bengaluru.

Teaching Experience

Aug 2025 – Nov 2025 **Teaching Assistant**, *Department of Physics, Indian Institute of Science*, Course: Mathematical Methods of Physics
Instructor: Prof. Justin David

Jan 2024 – April 2024 **Teaching Assistant**, *Department of Physics, Indian Institute of Science*, Course: Statistical Mechanics
Instructor: Prof. Aveek Bid

Reviewing Experience

2026 **Co-reviewer**, *for Communication Physics*, with Dr. Sumantra Sarkar

2025 **Co-reviewer**, *for Chaos*, with Dr. Sadhitro De

Awards, Honors and Grants

2026 **Kumari L A Meera Memorial Medal** for the Best Integrated PhD Student in Physical Sciences, Department of Physics, Indian Institute of Science

2026 **Travel Grant, from ANRF (Anusandhan National Research Foundation) ITS (International Travel Scheme)** - ITS/2026/003030, to attend *Active, Adaptive and Autonomous Matter*, France (06 July, 2026 to 18 July, 2026)

2024 **Best Teaching Assistant Award 2024**, Department of Physics, Indian Institute of Science

2023 **Professor M A Viswamitra Memorial Award** for the Best 1st Year Integrated PhD Student, Department of Physics, Indian Institute of Science

Co-curricular Activities

2026 - **Organizer**, GranSLAM (Granular, Soft, Living, Active Matter) Journal Club, Department of Physics, IISc